

Institute of Actuaries of India

**Subject ST7 – General Insurance :
Reserving & Capital Modelling**

May 2014 Examinations

INDICATIVE SOLUTIONS

Solution 1 :

Lines of business can be correlated because:

- They are impacted by similar events, e.g. a windstorm could impact both household and commercial property accounts
- Legal changes can affect several lines of business, e.g. a change in the discount rate required to calculate bodily injury awards would affect both employers' liability and motor classes
- The same claims team may handle claims from several lines of business and so changes to claims handling may impact on more than one line
- Problems with data may affect more than one line of business

Allowance for correlations in a stochastic model:

- A correlation matrix could be used
 - However this only allows for very simple dependencies between classes
- A copula is a more flexible approach, because it allows for multiple dependencies
 - For example, the Gumbel copula and the t-copula allow for increased correlations in the tails of the distribution
 - However, it can be difficult to parameterize dependencies from data, so judgment is important

The user must specify:

- underlying loss distributions for the classes of business
- a two-way correlation matrix between all distributions
- the form of the copula

[6 Marks]

Solution 2 :

- i. U/W results – treaty excess of loss reinsurance : will stabilize loss ratios because higher layers of loss will be ceded to the reinsurers ; advantage: only retain losses up to certain retention ; disadvantage: this reinsurance will not help if an increase in the frequency of smaller retained losses occurs .
- ii. Rapid growth – pro rata reinsurance: provides surplus relief which will lower the premium to surplus ratio and allow the company to continue growing; advantage: ceding commission reimburses primary company for their policy acquisition costs and provides surplus relief; disadvantage: cede premium on the profitable business, not just unprofitable business.
- iii. Higher policy limits – excess of loss reinsurance: provides large line capacity so the company can write higher policy limits than it could on its own; advantage: can write higher policy limits without retaining higher losses; disadvantage: can get expensive, and does not provide relief for an increase in frequency.
- iv. Higher volatile loss ratios - Stop Loss cover: this may protect the insurer from high volatility of loss ratios; advantage: can bring stability in claims experience till company gains experience of new product; disadvantage: may not be easily available and if available, may be very expensive

[8 Marks]

Solution 3 :

CMP Liability		
	Ceded%	Retained Loss
1	0%	10,000,000
2	33%	15,000,000
3	40%	45,000,000

CMP Property	
	Retained Loss
1	60,000,000
2	100,000,000
3	75,000,000
Others	3,000,000,000

Total retention	3,305,000,000
CAT recovery	2,700,000,000
Net Retention	605,000,000

[5 Marks]**Solution 4 :****i. Pattern of risk exposure**

Year 1	Year 2	Year 3	Year 4	Total
1	2	4	8	15

Proportion of earning in each year

Year 1	Year 2	Year 3	Year 4	Total
6.6667%		13.3333%		26.6667%
			53.3333%	100.0000%

UPR at the end of year 1, i.e. 31st March 2014 = 280,000 **[2]**

- ii.** To provide for deficiency in unearned premium reserve when UPR will not be able to cover expected loss & LAE, marginal expenses, DAC & policyholder dividends from the run-off of the remaining policy period **OR**
The purpose of an AURR (premium deficiency reserve) is to set up a reserve for the amount by which future loss and expenses attributed to unearned premiums exceed those unearned premiums. **[1]**

- iii.** Total loss expected from unexpired policy = $(97.5\% + 10\%) \times 280,000 = 301,000$. Hence Premium deficiency reserve = 21,000

Please note that fixed and general overhead expenses are not included in AURR calculation. Also, the reported loss will be covered by Loss Reserve rather than by AURR.

[3]**[Total Marks-6]**

Solution 5 :**i. 3 yr weighted average cumulative incurred loss LDF**

Age	12-24	24-36	36-48	48-60
LDF	1.0862	1.0914	1.0223	1.0000
Cum LDF	1.2119	1.1157	1.0223	1.0000

IBNR (Amounts in '000)

AY	Incurred to date	IBNR=Inc *(Cum LDF-1)
2009-10	24,760	0
2010-11	26,620	593
2011-12	28,310	3,276
2012-13	29,540	6,259
Total	109,230	10,129

[2]

ii.	LDF for 12-24 months	1.0862
	Claims incurred for AY 12-13	29540
	Expected IBNR emergence AY 12-13 in calendar 13-14	29540 * (1.0862-1)
		2545

[1]**iii. Paid Loss/Closed Claims (Amounts in '000)**

Paid severity	12	24	36	48	60
2008-09	110	114	118	122	138
2009-10	121	126	129	134	
2010-11	133	138	142		
2011-12	146	152			
2012-13	161				

Note: Calculated as (Incurred Losses-Case reserves)/Closed claims

Paid severity Trend

	12	24	36	48
2009-10	1.1	1.1	1.1	1.1
2010-11	1.1	1.1	1.1	
2011-12	1.1	1.1		
2012-13	1.1			

Note: Calculated as Paid severity for year x+1/Paid severity for year x
Loss Reserve by open claims (Amounts in '000)

Case reserve severity	12	24	36	48	60
2008-09	244	209	400	1,000	-
2009-10	269	229	442	1,100	
2010-11	296	252	484		
2011-12	407	351			
2012-13	448				

Trend of average case reserves

	12	24	36	48
2009-10	1.1	1.1	1.1	1.1
2010-11	1.1	1.1	1.1	

2011-12	1.4	1.4
2012-13	1.1	

Note: Calculated as case reserve severity for year x+1/Paid severity for year x.

It is observed from the data that there is a trend in the paid severity. However, since this trend is constant, this will get projected in future.

The average case reserves also had a smooth trend till AY 2010-11, but in 2011-12, all of a sudden the trend has changed with large strengthening of case reserves.

This can overestimate loss reserves.

[5]

[Total Marks-8]

Solution 6 :

(a) & (b):

Profit & Loss Account			
<i>Amounts in INR Mn</i>			
1	GWP		10000
2	Reinsurance (RI) ceded		-4000
3	NWP	(1+2)	6000
4	Change in UPR	explained below	-600 (a)
5	NEP	(3+4)	5400
6	Claims incurred	explained below	(3,240) (b)
7	Change in reserve	-300	Net of reinsurance
(Marks not deducted if 60% retention is not applied to prior years' liability increase of 500)			
8	Net Incurred Claims	(6+7)	(3,540)
9	Direct Commission Paid		-1500
10	RI comm. received		1000
11	Net Commission	(9+10)	-500
12	Net Insurance A/C	(5+8+11)	1,360
13	Management Expenses		-1500
14	Underwriting result	(12+13)	(140)
15	Investment Income		1,000
16	Insurance profit/loss	(14+15)	860
17	Provision for tax		(258)
18	Profit after tax	(16+17)	602
Net Loss Ratio		65.56%	Calculated as (8/5)
Combined Operating Ratio		102.59%	Calculated as (1-14/5)

[5]

Assumptions:

- a. Calculated on assumption that risks are written uniformly through the year
- b. Calculated on assumption that risks are written uniformly through the year. Also calculated as earned risks X frequency X severity X retention%
- c. Assumed that Incurred claims includes IBNR
- d. Assumed that there are no reinsurance bad debts.

[2]

[Total Marks-7]

Solution 7 :

- i. Reserving Risk:
- a. Mean = 100
 - b. Variance = $(\text{Mean} * \text{CV})^2 = (100 * 20\%)^2 = 400$
- Underwriting Risk:
- a. Mean = 50
 - b. Variance = $(\text{Mean} * \text{CV})^2 = (50 * 10\%)^2 = 25$
- Combined Risk:
- a. Mean = $100 + 50 = 150$
 - b. Variance = $400 + 25 + 2 * 0.30 * (400 * 25)^{0.5} = 425 + 2 * 0.30 * 20 * 5 = 485$
 - c. Coefficient of Variation = $(485^{0.5}) / 150 = 22\%$

Capital requirement (over and above the mean) at 95% level = $z\text{-value} * \text{SD}$
 $= 1.65 * \text{SQRT}(485) = 36.3.$

[3]

- ii. For OD, reserves constitute a smaller proportion of ultimate losses for past AYs.

So variation in these unpaid losses is not going to impact the estimate for future loss ratio as much as it'd do for TP.

General loss ratio assumptions for the future business are based to a greater extent for Third Party on the projected ultimate loss ratios for the past years; in other words, the estimates for unpaid losses.

Judicial precedents could impact both unpaid liabilities and future liabilities for TP.

So, greater correlation compared to OD in this aspect.

More correlation because of less years of experience and high growth, especially so for TP.

Because the assumptions in reserving and pricing are likely to be more interlinked.

[4]

[Total Marks-7]

Solution 8 :**i. Funds may be non-investible due to moneys held by:**

- Agents/brokers, who may pay premiums after holding with them for some time
- Policyholders', e.g.; installment premiums, end of year adjustment premiums
- Reinsurers, e.g.; delay in recoveries

The size of these non-investible funds may be influenced by:

- Mix of business
- Arrangements of premium receipt with other parties
- Debt collection efficiency

[3]**ii. Regulatory constraints**

The supervisory authorities may impose some restriction on the assets an insurer can hold and the way in which they are valued. This may arise in:

- The situation where the regulator deems it appropriate to impose restrictions on the market generally or on insurers in particular product lines or stages of development
- The situation where an insurer is in difficulty, having breached or being in danger of breaching regulations.

In general the regulator may impose following controls on the assets:

- Restriction on the amount of certain types of assets that can be taken into account when assessing solvency
- Custodianship of assets
- Prevention from holding certain assets
- Prescription to hold certain assets
- Requirement to hold mismatching reserves

External factors

- Tax treatment of different investments and the tax position of the general insurer
- Statutory, legal, ethical or voluntary restrictions on how the insurer may invest
- Statutory valuation requirements
- Solvency requirements – most territories have a step-wise process of intervention and therefore the strategy for maintaining solvency is targeted at a much higher level than the pure statutory minimum margin
- Rating agency constraints on capital required to maintain the insurer's desired rating
- Competition – strategy followed by other funds
- Regulatory constraints

[6]**[Total Marks-9]**

Solution 9 :

i.

Scenario	Initial Estimate Unpaid (at Year 0)	Retrospective Estimate of Unpaid at Year 0
1	82	86
2	90	90
3	98	94
4	92	96
5	100	100
6	108	104
7	100	105
8	110	110
9	120	115
Mean	100	100
Standard Deviation	10.83	8.90
Coefficient of Variation	10.83%	8.90%

- Variance calculated by treating it as a discrete distribution with equal probability for all scenarios

Retrospective Estimate of Unpaid at the end of Year 0 =
Paid During Year 1 + Case OS at the End of Year 1 + IBNR at the End of Year 1]

[4]

ii.

- Information that will be gained over one year: Projection of paid loss, reported loss, claims closed, claims reported, new judicial ruling, medical inflation etc. would also be required, along with inherent uncertainty.
- Methods: Standard methods to estimate reserve risk often take the ultimate-view. So need to use new methods that provide variability from a one-year view.
- Assumptions: Dependencies between lines of business and between risks or other assumptions could be different under a one-year view.

[3]

iii. Overall, less risk from a one-year view compared to the ultimate view.

Stability vs. responsiveness in reserving; reserves for long-tailed lines are not as responsive to incurred/paid movements as much as the reserves for short-tailed lines are. This is because stability is preferred over responsiveness under normal circumstances for long-tailed lines.

Usually little extra claim specific information is gained over a single year resulting in small changes to reserves using typical stochastic methods and consequently low measures of one-year reserve risk. For short-tailed lines, more information is gained in one-year.

One-year view would be closer to the ultimate view for a short-tailed line.

[3]**[Total Marks-10]**

Solution 10 :

- i. For each OD claim, obtain: accident date, reported date, make & model of the vehicle, City / Geography, Insured Declared Value (IDV) of the vehicle, NCB level, deductible level, type of accident (theft / fire / collision), more detailed type of accident (head-on collision, side-on collision etc.), garage details (network garage or other) etc.

Any other rating/risk factors and claim characteristics that have possible explanatory power of claims severity.

Total Paid till date along with the latest case reserves. Also, part payments / case reserve movements along with the dates, if possible.

Nil claims / re-opened claims information could also be obtained for a more detailed analysis.

[3]

- ii. The degree of detail (level of disaggregation) will depend on the credibility of available data and the characteristics found to have explanatory power during the factor selection for the model.

Fit a GLM or use any other technique to model the incurred claim amount based on its characteristics.

Response variable would be the total claim amount. Claim amount as a proportion of IDV could be more relevant for Theft etc.

Perform diagnostics to select a model. Check for reasonableness of the relativities for each factor.

Adjust fortrends, if any, based on Accident Year or Payment Year.

[4]

- iii. Severity is expected to increase with higher levels of NCB, everything else remaining the same.

This is because of Bonus Hunger, especially at higher NCB levels; smaller losses are not usually claimed by the insured to protect the NCB level.

For Theft claims, severity is not expected to vary as the claim amount would be the same as IDV.

[2]

[Total Marks-9]

Solution 11 :

- i. Rapid payout and no uncertainty in liabilities: Measurement of weather station data, with no field loss adjustment, allows rapid payouts. Also, there would not be any IBNR required under normal circumstances once the weather station data is available.

Transparency: Index insurance contracts usually allow the policyholder direct access to the Information on which the payouts will be calculated. Trust is strengthened by transparency.

No on-farm loss adjustment: On-farm loss adjustment is quite complex and costly and may not be credible in many low-income countries.

Lack of adverse selection and moral hazard: All producers within the defined area are treated equally; they have the same insurance payout conditions, regardless of their specific risk exposure. There is no benefit in individual producers trying to influence claims.

Low operational and transaction costs: Index insurance requires limited individual underwriting (client assessment). It can be distributed, and claims can be settled, at relatively lower cost.

Addresses correlated risks: Index products work best where there are correlated risks. With traditional products, perils such as drought are challenging to insurer. The product is most effective when targeted to protect against catastrophic weather events.

Reduce scope for fraudulent claims: As the claims payout is based on the observed values of the index, there is a reduced scope for fraud claims.

[5]

ii. and iii.

Basis risk: Basis risk is the risk that the loss experienced by the farmer could be different to the payout triggered. It could result in a farmer experiencing yield loss, but not receiving a payout, or in a payout being triggered without any loss being experienced.

There are various types of basis risk:

- Spatial basis risk: Local variations in the peril occurrence (e.g. rainfall) within the area surrounding a weather station.
- Temporal basis risk: Inter-annual variations in seasonal crop phases, meaning that the insurance phases are not temporally aligned with the intended crop growth stage.
- Product basis risk: Crop losses can be caused by many factors. Where there is no clear-cut relationship between loss and the indexed weather peril, basis risk can be high.

Reputational Risk: Poses a reputational risk for the insurer if the basis risk leads to a considerable gap between the payout and the actual loss on the ground.

Replication: The triggers, limits and increments of a specific product need to be adjusted to reflect the weather parameters of each weather station. Different product designs are required for different crop types (or at least generic crop types).

Technical capacity and expertise: Requires considerable technical work in its implementation and sustaining, particularly during the initial design phase for new products, in agro-meteorology and in operationalizing the products.

Lack of weather data: Availability and quality of weather data can drastically vary from region to region. Shortage of historical and real-time weather data is often a major hurdle.

Complexity & Selling Mechanism: Policies are difficult for farmers to understand. Requires effective organizational setup to market the product and gain sufficient volumes

Uncertainty in Weather Forecasts: Weather conditions are difficult to forecast and weather patterns are becoming more volatile and extreme events more frequent due to climate change

Correlation between risks: This product could be prone to accumulation of risk and an adverse weather event (for example, drought) could affect all risks in that region.

Reinsurance availability: Reinsurance may not be available and if available, could be expensive. Accumulation of risk means greater need for reinsurance as well.

Authenticity of Weather Data: Should be tamper-proof. However, this requirement is not often met.

[8]

[Total Marks-13]

Solution 12 :

- i. The underwriting cycle reflects the changes in premium rates and underwriting conditions as a result of changing levels of competition in the market.

The underwriting cycle results in market conditions fluctuating between a hard market and a soft market:

- hard market
 - limited capacity / supply means that premium rates are high, policy terms and conditions are more stringent and business is very profitable, leading to
 - more companies entering the market, capacity increasing so that the increase in competition leads to profit margins being squeezed until
- soft market
 - ... lots of capacity / supply means that rates are low, policy terms and conditions are less stringent and business is no longer profitable, leading to
 - ... some insurers leaving the market, reducing capacity and so the cycle continues.

[3]

- ii. a. During a hard-market period, the premium rates are high and the business is profitable. Therefore, the capital required should be lower as the chances of becoming insolvent due to inadequate premiums is lower.

However, the required capital would be more because of higher premiums.

- b. During a soft-market period, the premium rates are low and the business is not very profitable. Therefore, the capital required should be higher as the chances of becoming insolvent due to inadequate premiums are greater.

However, the required capital would be lower because of lower premiums.

[3]

- iii. It may not improve the effectiveness during a hard-market at all as claims, relative to premiums, from prior years are not expected to be very high.

It'd slightly help during a soft-market if the claims, relative to premiums, are high owing to low premium rate levels in the previous years.

However, if the company has been growing at a high rate, the capital requirement may still not be commensurate with risk.

[3]

- iv. Assumptions related to future premium rates or rate changes in the modelling of underwriting risk should take the period of UW cycle into consideration

More correlation from one year to the next w.r.t reserve risk, especially in the years are in the same period of the cycle.

Possible correlation between reserve risk and underwriting risk if the recent years and the future years in consideration are also in the same period of the cycle.

Possible correlations between LoBs, especially if they are in the same period of the UW Cycle. However, UW cycle period may vary from LoB to LoB.

[3]

[Total Marks-12]
